



Signs of life on Mars

NASA's Curiosity Rover found rock samples on Mars carbon, the element that indicates life at some point. <https://digit.in/july18-71-1>



Global warming on the Moon

NASA tapes indicate that humans may have been responsible for the Moon's temperature spike in the '70s <https://digit.in/july18-71-3>

Surgery in a pill

New drug could give the benefits of gastric bypass without the surgery

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According to the International Diabetes Federation, there were over 72 million cases of

diabetes in India in 2017. To put that into perspective, that number can contain the entire population of United Kingdom and still have enough room for Lebanon. Hold on, statistics can be twisted around to serve a purpose, right? So, ignore that number and look around you - how many people do you know who suffer from any type of diabetes? What we are trying to say is that diabetes is more than just an ailment in India, it is an epidemic.

Type-2 diabetes in particular is a highly prevalent form of diabetes where the body fails to produce enough insulin. One of the most effective ways to not only slow down its effects, but even cure it altogether is gastric bypass surgery. That surgery in India can cost anywhere between ₹4-7 lakh which is out of reach for a large part of the affected populace. Recently, researchers from Brigham and Women's Hospital in Boston, Massachusetts might have come up with a drug that removes the need for surgery altogether! Before we get to know more about that, we need to understand what gastric bypass surgery does.

GASTRIC BYPASS SURGERY

A gastric bypass can be done in a number of ways and can be used to

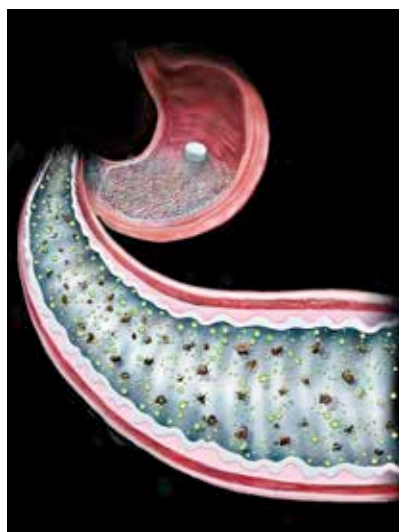


Illustration of the new pill developed at BWH that mimics the effects of gastric bypass surgery. Credit: Brigham and Women's Hospital and Randal McKenzie.

treat a number of conditions in the human body. Essentially, the surgery involves separating the stomach into a smaller 'upper' pouch and a much larger 'lower' remnant pouch and then connecting the small intestine to both. As a result of this, the subjects feel full with lesser food intake. Additionally, the food doesn't travel through the entire small intestine, hence, fewer nutrients are absorbed. Both of these lead to weight loss. So how exactly does gastric bypass surgery help treat diabetes?

The actual answer? No one knows. It could be one of two possibilities though. Essentially, hormones in our body that are gut related play an important role in controlling diabetes. Take insulin, for example. The production of insulin, and the regulation of blood sugar, is carried out by gut hormones. In a number of diabetic conditions, excess or lack of insulin is a primary cause. So, gastric bypass helps diabetics by changing the way their body produces gut hormones. It

could be because of the out-of-action stomach cavity or the disconnected section of the small intestine that stimulates these hormones. What it does lead to, according to numerous studies, is the reversal of diabetes in about 90 per cent of the patients.

THE PILL

The discovery of the Brigham and Women's Hospital researchers could bring the same benefits to people who cannot or do not opt for the surgery. Based on an FDA approved drug called sucralfate, their compound is called LuCI (Luminal Coating of the Intestine). It needs to be ingested before a meal. Once it reaches the stomach, it transforms into a sticky, paste like coating that sticks to the stomach wall and upper intestine. This prevents food contact with the bowel lining. Interestingly, the part of the intestine that does get coated is the part that is bypassed in gastric bypass surgery. The coating itself does not get absorbed and is eventually passed by the body.

This drug has been tested on rats so far, and has brought down glucose levels in their bodies by more than 40 per cent. Additionally, since it is based on an already-approved FDA drug, the compound itself shouldn't take long to be approved once it is ready to hit the stores. If it works on humans as well, it could be revolutionary in putting Type-2 diabetes into remission in cases where surgery is not an option. Since the effects of this compound are highly local, it is unlikely to have any serious side effects.

Regardless of this, the drug should not be expected to hit stores in the next few years. A time period of 5-10 years is more likely, but when it does become available, India could be one of its biggest beneficiaries. 